Justin Fox

Data Engineer

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SUMMARY

Data Engineer with 10+ years of experience in data systems, analytics engineering, and automation, including 3+ years building scalable ETL/ELT pipelines, cloud data platforms, and warehouse architectures. Experienced in designing end-to-end data workflows using Python, SQL, dbt, and modern cloud platforms (AWS, GCP, Snowflake, Databricks), with a strong focus on data reliability, scalability, and self-service analytics enablement.

SKILLS

Languages & Frameworks: SQL, Python, dbt, PySpark, Pandas, Scikit-learn, Requests

Data Engineering & Analytics: ETL/ELT pipelines, Delta Lake, API integration, Batch/Stream processing, Data modeling, NoSQL, A/B testing, Machine learning, Data validation

Tools & Platforms: AWS (S3, Lambda), GCP (Cloud Run, Pub/Sub, Firestore, BigQuery), Snowflake, Databricks, Power BI

EXPERIENCE

Data Strategy & Analytics Lead (Volunteer) | Jesus Said Go | Remote | August 2025 - Present

- Developed CRM automation systems using Zoho Deluge and API integrations, including a scheduled reporting function that compiles performance metrics and delivers automated insights to users via WhatsApp API, as well as a JSON webhook API to receive and validate structured ingestion of transcribed voice-call reports into CRM workflows.
- Redesigned and optimized end-to-end analytics workflows in Zoho Analytics, including data transformation for accuracy, automated reporting schedules, and a public-facing dashboard to improve stakeholder visibility and data accessibility.

Principal Business Analyst / Data Administrator | San Joaquin County Probation | Stockton, CA | March 2023 - Present

- Built an automated data orchestration system using Python and SQL, running 30+ scheduled ETL jobs to ingest, transform, and load multi-source data into a centralized SQL Server data warehouse supporting 60+ tables across 4 schemas.
- Led and mentored a team of analysts while designing and delivering 20+ Power BI dashboards built on optimized SQL data models and warehouse datasets, improving reporting efficiency and enabling scalable, self-service analytics.

Business Analyst II | San Joaquin County Probation | Stockton, CA | December 2014 – March 2023

- Automated 20+ recurring reports and data extraction processes using Python, SQL, and VBA, reducing manual workload and improving accuracy.
- Developed an NLP-based machine learning model using probation case notes to predict client outcomes, achieving 89% accuracy and delivering actionable insights for management decision-making.

PROJECTS

Etsy Shop ELT Pipeline | [Github]

- Designed and deployed a fully automated cloud-native ELT pipeline that extracts incremental sales and fulfillment data from Etsy and Gelato APIs via AWS Lambda, stores raw JSON in partitioned Amazon S3, ingests into Snowflake using Snowpipe, and transforms it with dbt into a dimensional star schema with automated data quality tests and scheduled warehouse builds via Snowflake Tasks.

Website Clickstream Pipeline & Analytics | [Github]

- Designed and implemented, for a live production website for a nonprofit, a real-time clickstream analytics pipeline using JavaScript, Google Cloud Run, Pub/Sub, Firestore, BigQuery, and Looker to capture anonymized website engagement events and deliver session-level analytics.

Earthquake ETL Lakehouse Pipeline | [Github]

- An end-to-end PySpark + Delta Lake pipeline in Databricks ingesting real-time USGS earthquake API data into Bronze/Silver/Gold medallion layers with metadata logging, optimized analytic tables, and visualization outputs for reporting.

EDUCATION

California State University - Stanislaus

B.S. Mathematics | Minor in Computer Science | December 2013